

What is Food?

A Method for Evaluating Substances by Body Response

Registry: [[HOWL-BODY-2-2026](#)]

Series Path: [[HOWL-BODY-1-2026](#)] → [[HOWL-BODY-2-2026](#)]

Date: April 2026

DOI: 10.5281/zenodo.19719331

Domain: Body and Biology

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Abstract

Food is a substance that stabilizes the body's homeostasis machine when ingested. Not-food is a substance that destabilizes it. Most substances do both in varying proportions, and the category of any given substance depends on the net regulatory effect over the intake pattern. This paper teaches the method for evaluating any substance against the question "is this food?" The method rests on the body as a continuously regulating system; food as an input that either helps the regulation succeed or taxes it; and the operational split between regular food (what the body sustains on as ongoing input) and irregular food (what the body handles occasionally without lasting harm). The paper is scoped to food in its relation to the body. Cultural framing, meal sociology, nutrition prescriptions, and medical guidance are out of scope. The reader finishes able to apply a four-question decision rule to any substance and place it in a category that corresponds to what the body will actually do with it.

1. What You Are Reading This For

You eat. You have eaten every day of your life and will continue to eat for the rest of it. Every substance you ingest produces a response in your body, and the accumulated responses over decades are most of what determines your health trajectory. Despite this, you probably don't have a stable answer to the question "is this food?" for most of what you eat.

Your current answer, if you have one, is probably cultural. Food is what's in the grocery store. Food is what's on menus. Food is what people eat at meals. This answer collapses on the first hard case. Is a bag of chips food? Is cake food? Is a can of soda food? The cultural answer gives you "yes" for all three, because these substances are sold, eaten, and marketed as food. The body's answer is different, and the difference matters for your life.

This paper replaces the cultural answer with a body-response answer. After reading it, you will be able to ask “is this food?” of any substance and produce a stable answer that corresponds to what your body will actually do with it over time. The answer will not always be yes or no. For many substances, the answer will be “food at this frequency, not food at that frequency,” or “food in combination, not food alone.” This isn’t evasion — it’s precision. The body’s response is pattern-dependent, and the answer has to track the body.

If you finish this paper and still want a single yes/no label per substance without considering pattern, you haven’t crossed the frame. The disciplines that follow will feel like unnecessary complications, and you will keep treating substances as food or not-food based on cultural category rather than what your body does with them.

2. The Three Inputs to Human Life

Humans require three inputs to stay alive: air to breathe, water to drink, food to eat. Other activities — sleep, movement — are required for life but are not inputs in the same sense. Other inputs exist — light, for example, drives discovery — but are not the subject of this paper.

The three inputs have different tolerance timescales. Without air, death arrives in minutes. Without water, days. Without food, weeks. The order reflects the biological urgency of each input, and the urgency explains something about the body: the systems that depend on air fail fastest, the systems that depend on water fail next, and the systems that depend on food fail slowest. All three are required. None is optional. The time to failure tells you which is running closest to the edge at any moment.

Air and water are named to establish the context for food, and then set aside. The paper’s subject is food — the input with the most variation in what counts, the most cultural confusion, and the most decisions the reader makes daily. Food is the input you reach for several times a day, choosing from many options, and the choices compound over a lifetime. Air and water are mostly not a matter of choice. Food is.

The paper’s frame is set by this: food is one of three inputs to a biological system that runs continuously. It is not a cultural activity, an aesthetic experience, or a social event, though it intersects with all of those. At the body-response level, food is a substance that enters the system and changes the system’s regulatory state.

3. What Food Is Not

Before defining food, the paper clears away the non-food categories. These are substances that can be ingested, that affect the body, but that do not function as food. Confusing any of them with food collapses the question “is this food?” into a definition too broad to be useful.

Water is not food. Water is essential. Water is ingested. A person who drinks adequate water but eats nothing dies of starvation in approximately fifty days, and weakens considerably sooner. Water prevents dehydration; it does not prevent starvation. These

are different mortality paths. A definition of food that includes water as food cannot distinguish them. Water is a non-food essential.

Water has a lethal dose and is not poison. Enough water in a short enough window kills through hyponatremia and related mechanisms. The mortality is real. Water is not attacking the body; the body's electrolyte regulation is overwhelmed by the water load. This is homeostasis failure, not chemical injury. Substances that kill by overwhelming regulatory capacity without being chemically hostile belong in their own category: homeostasis-disruptors at excess dose. Salt at extreme intake and oxygen at high pressure are other examples. None of these are poisons; none are food.

Poisons disrupt the body at any dose. Cyanide, arsenic, botulinum toxin. No intake level helps the body sustain function. Every dose harms, with the magnitude of harm scaling with dose. Poisons have no safe range, no required range. Their relationship to the body is purely disruptive. Substances with dose-dependent safety profiles — safe or required in some range, harmful above it — are not poisons in this sense, even if they have LD50s.

Medicine is not food. Medicine intervenes in body function when function has departed from normal. The intervention produces a specific effect for a limited purpose. Food sustains normal function by providing materials the body incorporates into ongoing operation. These are different relationships to the body.

Medicine is a situational category, not a substance category. A substance functions as medicine only in the intersection of correct dose and correct indication. Outside that intersection, the same substance functions as poison, inert exposure, or something else. An NSAID taken for inflammation is medicine. An NSAID taken without inflammation is the same mechanism running uselessly through a body that didn't need it, producing all the downsides of the mechanism with no therapeutic return. The mechanism doesn't know whether it's wanted.

All medicines become poisons at incorrect dose or without indication. This is not a failure of medicine; it is a clarification of what medicine is. The substance is medicine when the intervention is wanted and the dose is therapeutic. Otherwise, it is the substance acting in a body without justification for its action.

The categories:

- Poisons: harm at any dose.
- Non-food essentials: required for life, ingested or inhaled, not food. Water, air, salt.
- Homeostasis-disruptors at excess dose: safe in range, lethal beyond. Water past LD50, oxygen at high pressure, sodium at extreme intake.
- Medicines: situational interventions; function as medicine only at correct dose with correct indication.

Food is none of these. Food occupies a distinct category, and the definition emerges from what food does that none of these non-food categories do.

4. The Body as a Homeostasis Machine

The body is a homeostasis machine. It is running continuous regulation on hundreds of parameters simultaneously, consuming energy to keep the body inside its viable envelope. Blood glucose in range. Blood pressure in range. Core temperature in range. Electrolyte concentrations in range. Blood pH in range. Hydration in range. Hormones in range. Countless others.

The ranges are not static. They shift with activity, time of day, season, age. But they are maintained within tolerances by regulatory systems that run actively, at all times, whether or not the person notices. Every heartbeat, every breath, every digestive contraction, every nerve firing, every enzyme produced is part of the ongoing work of staying in range.

The default state of the body is not rest. The default state is active regulation. The body is always working to stay alive, because staying alive requires continuous adjustment against the disturbances that would push the body out of range — heat, cold, exertion, stress, infection, injury, and above all, the intake of substances from outside.

Food enters this machine. Whatever food does, it does in the context of this regulation. The right question about any ingested substance is not “what is it made of?” — that is chemistry, and chemistry is upstream of the question that matters. The right question is: what does this substance do to the machine?

The chemistry is a guide to predicting the response. The response is what determines the substance’s status. A substance is food to the extent that the machine can run on it. A substance is not-food to the extent that it taxes the machine’s regulation or fails to supply what the regulation needs.

This reframes the evaluation. Instead of asking whether a substance has the right nutrients, or whether it’s natural, or whether experts approve of it, the reader asks: how does my body respond to this substance, across the patterns of intake that would be realistic for me? The answer is observable, because the body keeps responding, and the response accumulates over years into measurable health outcomes.

5. The Operative Definition

Food is a substance that stabilizes the homeostasis machine when ingested. Not-food is a substance that destabilizes it. Most substances do both in varying proportions. The category of any given substance depends on the net regulatory effect over the intake pattern.

Each word of this definition is load-bearing.

Stabilizes means providing materials the regulatory systems use, within a load the systems can handle. Energy substrate for running the body’s processes. Structural materials for tissue maintenance and repair. Regulatory materials — vitamins, minerals, specific compounds — that the regulatory systems require as cofactors, substrates, or signaling components. When a substance provides these in usable form and within manageable quantities, the body runs its regulation with less internal cost. Homeostasis is maintained more easily.

Destabilizes means failing to provide needed materials, overloading specific regulatory systems, or adding clearance work without contributing useful content. The body has to work harder to stay in range. Regulatory systems run longer or at higher intensity, or in compensation modes that weren't designed for sustained operation. Over time, regulatory capacity degrades. Ranges that were being maintained start to drift. Parameters that were in balance become imbalanced. Disease is the name for the accumulated consequence of regulatory systems failing to maintain range.

In varying proportions acknowledges that few substances are purely one or the other. A substance can stabilize on some axes (provide energy, contribute some micronutrients) and destabilize on others (require clearance, strain specific regulatory pathways). The reader should not expect a clean binary at the substance level. The binary emerges from the net effect over the intake pattern.

Net regulatory effect over the intake pattern is where the category is determined. A substance eaten occasionally produces a transient regulatory perturbation that the body absorbs and clears, with baseline restored quickly. The same substance eaten frequently keeps the perturbation ongoing, and the regulatory systems begin to accumulate wear. The substance hasn't changed. Its relationship to the body has, because the pattern changed.

Food, then, is not a property of substances in isolation. It is a property of relationships between substances and bodies over time. The same substance can be food for one person at one intake pattern and not-food for another person or at another intake pattern. This is not relativism. It is precision about where the category lives.

6. The Sustaining Gradient

Within the category of food, substances vary in how completely they sustain the body as sole intake. A substance that provides most of what the body needs can sustain a person indefinitely when eaten alone. A substance that provides some of what the body needs can contribute to sustenance only as part of a larger intake. A substance that provides little of what the body needs cannot sustain life even when it is the only thing available.

Insufficient substances. Pepper seeds can be ingested. They are not toxic in reasonable amounts. They contain some trace organic matter. A person eating only pepper seeds dies on approximately the starvation timeline, because the substance provides so little that the body cannot sustain on it. The pepper seeds are functionally equivalent to no food, at least in the mortality sense. Many common substances are in this category: things the body can ingest without harm but cannot extract meaningful sustenance from.

Partial food, acutely non-sustaining. Rabbit meat is food. Protein, water content, some vitamins and minerals, digestible, historically eaten by many populations. But a person eating only rabbit meat dies in weeks. The mechanism is specific: rabbit is extremely lean, and the body cannot metabolize enough protein to meet caloric needs without exceeding the liver's urea clearance capacity. Nitrogen accumulates. The regulatory system for nitrogen balance fails. The person dies faster than pure starvation, actively poisoned by their own metabolic waste. The food is real. The food fails as sole intake.

Partial food, acutely sustaining, chronically insufficient. Bread with water is the

historical example. Bread provides calories, and a person on bread and water survives in the short term. But bread is missing essential fatty acids, missing vitamin C, missing B12, missing complete amino acid profile, missing much of what the body needs for sustained regulation. Deficiency diseases develop over months to years. The person doesn't die of starvation. They die of deficiency, or become chronically ill, or survive in diminished health. "Bread and water" as historical punishment rations worked precisely because bread occupies this category — enough to prevent acute starvation, not enough to preserve health.

Sustaining food with specific chronic costs. Pig meat alone is broadly sustaining. Protein, fat, water, and enough micronutrients to avoid major acute deficiencies. A person eating only pig meat can live for a long time. They may develop gout from purine load, cardiovascular effects from high saturated fat, vitamin C deficiency over years if they eat no organ meats. These are real costs but they are slow, and the person doesn't die the way a rabbit-only or bread-only eater eventually does.

Broadly sustaining food in combination. A diversified diet across varied whole and lightly processed foods produces net stability across most regulatory axes. Individual tolerance varies — some people don't handle dairy, some have specific intolerances — but the core pattern of varied real food is what the human body has sustained on across evolutionary time.

The fully sustaining single substance mostly doesn't exist. No single natural food provides everything the body needs for a lifetime of sole intake without developing problems. Certain combinations do. Certain specifically formulated products aim at this target. But the question "what single food could you live on indefinitely in full health?" has effectively no answer, and this is a feature of the body's design — the regulatory systems require varied inputs because they evolved in environments where inputs were varied.

The gradient matters because it clarifies that "is this food?" sometimes resolves to "partial food — sustaining only in combination, not alone." This is still food, but with a qualification. A person who treats pepper seeds as partial food the way they treat broccoli as partial food is misclassifying both. Pepper seeds are below the threshold of partial food. Broccoli is genuinely partial food — it contributes real nutrition, and it belongs in the sustaining diet, but it cannot be eaten alone.

7. The Duration Axis — Acute and Chronic Food Status

Some substances are food on short timescales and not-food on long ones. This is the time-dependent food status, and it produces a category that the sustaining gradient alone cannot capture.

Processed flour is the clearest case. A person eating processed flour for two weeks during a food-scarce situation survives. The body extracts calories. Acute deficiency disease doesn't develop in that window. The flour sustained the person through the short-term need. In that context, processed flour functioned as food.

A person eating processed flour as a substantial part of their diet for twenty years produces a different body response. Blood glucose regulation strains from repeated high-glycemic

load. Insulin sensitivity degrades. The pancreas wears. Micronutrient-dependent systems fail as the flour fails to supply what whole grains would have supplied. Gut regulation shifts. Cardiovascular risk rises. The body is wearing down under sustained exposure to a substance that provides calories without contributing much else, while demanding regulatory work that exceeds what the input justifies.

The substance hasn't changed. The body's response has, because the exposure duration changed. At brief exposure, the regulatory systems absorb the input and recover. At sustained exposure, the systems run under chronic load and degrade.

This produces a real category: short-term food, long-term poison. The substance passes the acute food test (provides calories, doesn't acutely kill) and fails the chronic food test (produces measurable regulatory damage under sustained intake). The body's response is time-dependent, and the substance's category depends on which timescale the question is being asked at.

Most processed foods in the modern diet occupy this category. They pass the acute test. They fail the chronic test. They function as food when eaten briefly and as slow damage when eaten consistently. A person who eats processed food for a week while traveling is using it as food. A person who eats processed food as daily intake for decades is being slowly worn down by it, while still being fed by it in the acute sense.

This is why "is this food?" sometimes requires "for how long?" as part of the answer. Without duration, the question is ambiguous for substances in this category. A person asking whether processed flour is food cannot get a stable answer without specifying the intake pattern. At two weeks, yes. At twenty years of daily intake, no — at that pattern, it is a chronic destabilizer wearing the regulatory machine down.

8. The Sliding Scale Within Food Names

Common food names cover ranges of substances, not single substances. "Bread" covers home-made whole wheat through cheap commercial white bread. "Meat" covers grass-fed whole cuts through heavily processed meat products. "Milk" covers raw whole milk through ultra-pasteurized shelf-stable milk with additives. "Vegetables" covers garden-grown produce through canned products with added sodium and preservatives.

The body responds to the specific substance, not to the name. A definition of food that works at the category level — "is bread food?" — fails because the category contains substances with different answers.

Bread across its range.

Home-made whole wheat bread, made from fresh-milled flour, water, salt, and leavening, retains most of what whole wheat contains: B vitamins, vitamin E, magnesium, zinc, iron, selenium, fiber, and the essential fatty acids from the germ. A person eating this bread as part of a varied diet gets real nutrition from it. Eaten alone, it would produce deficiency over years rather than months, and more slowly than stripped white bread would.

Store-bought whole wheat bread typically blends whole wheat flour with refined flour, preservatives, dough conditioners, emulsifiers, added sugar, added vital wheat gluten, sig-

nificant sodium, and sometimes industrial vegetable oils. The nutritional profile is lower than home-made. The additive load adds clearance work. At sustained intake, the body processes this as a degraded version of bread — still partial food, closer to the destabilizing end.

White bread is made from refined flour. The bran and germ are removed, taking with them most vitamins, most minerals, most fiber, and the germ fats. Enrichment adds back a specific subset — thiamine, riboflavin, niacin, folic acid, iron — to prevent the acute deficiency diseases that unenriched white bread historically produced. The enrichment does not restore the full profile. Vitamin E, B6, magnesium, zinc, selenium, fiber, and germ fats are not replaced. A person eating only white bread develops deficiency faster than on whole wheat, though acute deficiency diseases prevented by enrichment don't appear.

Cheap white bread takes cost reduction further. Lowest-grade refined flour, minimum regulatory enrichment, cheap vegetable oils, high-fructose corn syrup, maximum use of conditioners and preservatives, and artificial softeners to imitate freshness. The caloric content is intact. The nutritional profile is at the regulatory floor. The additive load is highest. Sustained intake produces deficiency on a faster timeline and accumulates exposure to specific additives with their own costs.

These are four different substances by body response. Treating them as the same substance because they share a name obscures what the body is actually doing with each. A person who eats home-made whole wheat bread sometimes, as part of a varied diet, is eating real food. A person who eats cheap white bread three meals a day is being slowly worn down, with the substance contributing both the deficiency and the additive load simultaneously.

The principle is general. Most food categories span a range like this. “Meat” as grass-fed beef with the animal's natural fat profile is very different from “meat” as processed meat product with binders, fillers, nitrites, and vegetable protein extensions. “Milk” as whole milk from pastured animals is different from reconstituted milk powder with added vitamins. “Vegetables” as fresh produce is different from canned produce with added sodium and preservatives. The names are the same. The substances are not.

To evaluate whether something is food, the reader has to look past the name. The question is applied to the specific substance, not the category the substance is commonly labeled under.

9. Regular Food and Irregular Food

The practical split. Food divides into what the body can sustain on as ongoing input and what the body handles occasionally without lasting harm. These are the two categories the reader will use for intake decisions.

Regular food is what the homeostasis machine can run on indefinitely. The body handles regular food as ongoing input without regulatory wear. The systems designed to process food do their work, extract what's needed, clear what isn't, and return to baseline ready for the next intake. There's no accumulating load, no progressive drift in any regulatory parameter, no specific system getting worn down by repeated exposure.

Regular food includes meats in reasonable variety, eggs, dairy for those who tolerate it, vegetables, fruits as whole foods, whole grains in moderation, legumes, nuts, seeds, fats from whole-food sources, water. Individual tolerance varies — some people don't handle dairy, some don't handle certain grains, some have specific intolerances — but the core is foods the human body has eaten across evolutionary time and handles without chronic destabilization.

Regular food isn't defined by being "healthy" in any fashionable sense. It's defined by whether the body can sustain on it as ongoing input. A person eating regular food, across years and decades, stays in homeostatic range. The machine runs, the regulation succeeds, health is preserved as a downstream consequence.

Irregular food is what the body handles as a periodic input but not as ongoing intake. A portion, a meal, an occasion — the body absorbs the load, processes it, compensates for the transient destabilization, returns to baseline. The regulatory systems have enough capacity to handle occasional perturbations. No lasting damage from infrequent exposure.

The same substance as frequent or daily input is destabilizing. The regulatory systems don't get time to recover between loads. Destabilization accumulates. Regulatory capacity degrades. The body moves from handling it to being worn by it.

Irregular food includes cake, chips, pastries, sweetened drinks, fried foods, heavily processed snacks, alcohol, ice cream, and concentrated sugar in most forms. None of these can be regular intake without destabilization. All of them can be occasional intake without lasting damage.

The split is defined by body response, not by cultural category. Some substances treated as regular food culturally are actually irregular food by body response. Breakfast cereals marketed as healthy often have sugar content closer to cake than to bread. Flavored yogurts frequently contain enough added sugar to function as irregular food. Fruit juices deliver sugar loads the body handles differently from whole fruit. These are category errors at the cultural level; the body responds to the substances as the substances are.

Some substances sit near the boundary. White bread is closer to regular than cake is; both are processed, but at equivalent intake frequency, cake destabilizes faster because cake concentrates more destabilizing components. Home-made whole wheat is closer to regular than store-bought whole wheat. The gradient within irregular food matters because it tells the reader how often is too often for a given substance. White bread several times a week is further toward destabilization than white bread twice a month. Cake daily is severely destabilizing. Cake at a birthday is absorbed and cleared.

The split gives the reader an operational rule. For any substance, ask: can the body sustain on this as ongoing intake, or is this something the body handles occasionally but wears under as frequent intake? If sustained is fine, it's regular food — include as ongoing diet. If sustained destabilizes, it's irregular food — include occasionally, if at all, with awareness that frequency determines whether it functions as food.

10. Sugar and Salt — Two Specific Cases

Sugar and salt are worth examining separately because they land in different places on the spectrum despite being commonly paired.

Salt is not food. Salt cannot sustain life as sole intake. A person eating only salt dies — first from dehydration, because the osmotic load pulls water out of cells and into the gut to dilute the salt for excretion, and the kidneys cannot keep up with clearing it. They die faster than water-only starvation, not slower, because salt actively destabilizes homeostasis while providing no energy to run the body.

Salt's role in the body is regulatory. Sodium and chloride are required for nerve function, muscle contraction, fluid balance, blood pressure regulation. The body needs a certain intake range. Below that range, regulation fails. Above that range, regulation is strained. But salt isn't providing substrate for the body's ongoing function. It's providing electrolytes the regulatory systems use while doing their work. The systems need sodium to operate; sodium is not fuel.

Salt belongs in the non-food essentials category with water and air. Required for life, ingested, not food.

Sugar is food, acutely, and is destabilizing on almost every longer scale. Sugar is pure caloric substrate. The body can extract energy from it directly. A person eating only sugar and water gets calories. They don't starve on the rabbit-meat timescale. They don't die in days the way a salt-only person does.

But the body's response to sugar-only intake destabilizes across nearly every regulatory system faster and harder than any of the other “bad food” cases. Blood glucose regulation fails fast as insulin production runs at redline and insulin sensitivity degrades. Protein regulation fails as the body catabolizes its own tissue for amino acids. Fat regulation fails as essential fatty acid supplies run out. Micronutrient regulation fails across the board because pure sugar has essentially none. Gut regulation fails as the microbiome shifts. Mineral regulation fails as sugar metabolism consumes minerals with no replacement.

A person on sugar-only survives longer than rabbit-only (no nitrogen toxicity) and longer than salt-only (calories are available), but the compound regulatory failures reach lethality on the scale of weeks to a few months. Sugar is the most destabilizing of the “can technically sustain life briefly” category. It is closer to the poison end of the food spectrum than most things people consider food.

At occasional intake, sugar is absorbed and cleared — the transient spike is handled, and the body returns to baseline. At sustained intake, sugar wears the regulatory machine faster than most other inputs. This makes sugar firmly irregular food at any meaningful quantity, and substances that concentrate sugar — candy, sweetened drinks, many desserts — inherit this position.

11. Cake as a Composite

Cake is a composite of substances we've already placed on the spectrum. Processed flour, sugar, fat, eggs, sometimes milk, flavoring. The dominant components by mass are usu-

ally flour and sugar, with fat as the third major component. The body responds to the combination, and the response depends on what the components are, in what proportions, and at what frequency.

The egg and milk content contributes real nutrition — complete protein, fat-soluble vitamins, minerals. The fat contributes energy and, depending on the fat, either useful materials (butter, with its fat-soluble vitamins) or clearance work (industrial shortening or partially hydrogenated oil). The flour adds processed carbohydrate with the sustained load of processed flour. The sugar dominates the destabilization profile, because sugar at this concentration delivers a glycemic load higher than most people would eat in pure sugar form — the combination with fat and flour makes the sugar palatable in quantities that would be unpalatable as pure sugar.

Cake at occasional intake — a slice at a birthday, rarely — is a periodic load the homeostasis machine handles. The sugar spike is real but transient. The flour adds to the load briefly. The fat slows absorption slightly. The small amount of real nutrition from eggs and milk contributes. For an occasional input, the destabilization is within the machine's compensation capacity, and the machine handles it and returns to baseline.

Cake as a frequent part of intake is a different response. The sugar load is repeated, and insulin regulation strains. The flour adds chronic load on the same regulatory system. The fat, if it's industrial, adds clearance work. The small amount of real nutrition doesn't compensate for the magnitude of the destabilizing load. Over time, the body's regulatory capacity degrades the same way it does for sustained chip or bread-dominated intake, with the sugar component amplified.

Cake as a regular daily food is destabilizing across most systems, producing the chronic disease profile characteristic of high-sugar, processed-flour-dominated intake. The person survives, because calories are available and some nutrition is present, but the regulatory machine wears down faster than for most single-ingredient irregular foods.

So: is cake food? At occasional intake, cake functions as food — the body absorbs it, uses what it can, clears the rest, and returns to baseline without lasting cost. At frequent or daily intake, cake is not functioning as food. It's a chronic destabilizing input that wears the regulatory machine faster than most other categories because it concentrates sugar and processed flour.

The answer to “is cake food?” requires the intake context: at what frequency, as part of what overall pattern, for how long? Without the context, the question is malformed for this class of substance.

12. The Decision Rule

The frame now resolves into an actionable rule the reader can apply to any substance. This is the method the paper is installing.

For any substance the reader is considering eating, ask:

First: what does this substance do for the homeostasis machine? Does it provide energy substrate the body can use? Does it provide structural materials — amino acids,

essential fatty acids, minerals for tissue? Does it provide regulatory materials — vitamins, minerals as cofactors, compounds the body can't synthesize itself? The more completely a substance provides these, the more it stabilizes. A substance that provides only calories (pure sugar) is far less stabilizing than one that provides calories plus protein plus fat plus micronutrients (eggs).

Second: what load does this substance put on the regulatory systems? Does it sit within normal capacity, or does it strain specific systems? Excess sugar strains insulin and glucose regulation. Excess protein strains nitrogen clearance. Excess salt strains fluid and blood pressure regulation. Oxidized fats add clearance work. Additives the body must process without benefit add load. The more load a substance imposes without providing compensating materials, the more it destabilizes.

Third: what happens under sustained intake? If this substance were a regular part of intake, over months and years, does the body sustain, or does regulatory capacity degrade? The question is about the trajectory, not the single meal. Substances that pass the first two questions at occasional intake but fail under sustained intake are the short-term-food, long-term-poison category. This is where most processed foods live.

Fourth: what happens under occasional intake? Some substances that cannot be sustained on can still be eaten occasionally without lasting damage. The body absorbs the perturbation, recovers, and returns to baseline. This is what separates irregular food from not-food. Irregular food is handled occasionally. Not-food (poison, homeostasis-disruptor) causes damage even at occasional intake.

Based on these four questions, the substance falls into a category:

- **Not food — poison:** harm at any dose. Cyanide, arsenic, certain industrial contaminants. Avoid.
- **Not food — non-food essential:** required for regulatory function, cannot sustain life. Water, salt, air. Use as needed; don't mistake for food.
- **Not food — homeostasis-disruptor at excess dose:** safe in range, lethal beyond. Stay in range.
- **Not food — insufficient substance:** ingestible but provides too little to contribute. Pepper seeds and similar. Not worth eating as food.
- **Medicine:** situational intervention. Use when indicated, at correct dose, for limited time.
- **Regular food:** body sustains as ongoing input. Eat as the basis of diet.
- **Irregular food:** body handles occasionally. Eat infrequently if at all; treat frequency as part of what determines its effect.
- **Short-term food, long-term poison:** passes acute test, fails chronic test. Recognize as irregular food and hold to genuinely occasional intake, or avoid.

The answer depends on intake pattern. The same substance can resolve into different categories at different patterns. This is not a failure of the rule — it is precision about

where the category lives.

13. Applying the Rule to the Paper's Cases

The reader's practice run. Walk through the cases the paper has covered, showing how the rule resolves each.

Pepper seeds. Step 1: provides essentially nothing. Step 2: minimal load, but the point is moot. Step 3: would fail within weeks of sole intake through insufficiency. Step 4: no damage at occasional intake, but no benefit either. Category: not food — insufficient substance.

Water. Step 1: no substrate, provides hydration the regulatory systems require. Step 2: load within range; lethal above. Step 3: sustains regulation but not body mass. Step 4: handled. Category: not food — non-food essential.

Salt. Step 1: provides electrolytes the regulatory systems use, no substrate. Step 2: required in range, strains systems above. Step 3: lethal at sole intake through dehydration. Step 4: required periodically for regulation. Category: not food — non-food essential.

Cyanide. Step 1: disrupts cellular respiration. Step 2: catastrophic load at any dose. Step 3 and 4 moot. Category: not food — poison.

Rabbit meat as sole intake. Step 1: protein, some nutrients, insufficient fat. Step 2: nitrogen clearance overloaded at quantities needed for calories. Step 3: destabilizes within weeks at sole intake. Step 4: fine at occasional intake as part of larger diet. Category: partial food, sustaining only in combination. Regular food in combination; not sustainable alone.

Pig meat. Step 1: protein, fat, water, micronutrients. Step 2: manageable load. Step 3: sustains for long periods with slowly developing specific problems. Step 4: absorbed easily. Category: regular food, with specific chronic costs at high fraction of intake.

Home-made whole wheat bread. Step 1: carbohydrate, some protein, fiber, real micronutrient content. Step 2: manageable load. Step 3: partial — missing essential nutrients for sole intake, but fine in combination. Step 4: absorbed easily. Category: regular food as part of varied diet.

White bread. Step 1: carbohydrate, minimum-enrichment nutrients, little else. Step 2: moderate glycemic load. Step 3: chronic insufficiency if dominant. Step 4: absorbed easily. Category: between regular and irregular; regular food at small fractions of varied intake, irregular if frequent or dominant.

Cheap white bread. As white bread, plus additive load and worse fat profile. Further toward irregular than white bread. Dominant intake is destabilizing.

Cheap potato chips. Step 1: calories from oil and starch; minimal other content. Step 2: oxidized fat clearance, high sodium, engineered hyperpalatability disrupting appetite regulation. Step 3: chronic destabilization across multiple systems. Step 4: absorbed at occasional intake. Category: irregular food, further toward destabilization than cheap white bread.

Pure sugar. Step 1: calories only. Step 2: severe load on glucose and insulin regulation. Step 3: broad regulatory failure within weeks at sole intake; chronic destabilization at sustained high intake. Step 4: absorbed at occasional small quantities. Category: irregular food at most, closer to poison than most foods.

Cake. Step 1: small real nutrition from eggs and milk; dominant calories from sugar and flour. Step 2: severe sugar load concentrated with flour load. Step 3: destabilizes fast at frequent intake. Step 4: absorbed at occasional intake. Category: irregular food; frequency determines whether it functions as food or slow destabilizer.

The reader can reproduce this reasoning for any substance. The placements are defensible, and where the reader disagrees with a specific placement, the rule gives them the basis for the disagreement, which means the method has transferred.

14. The Ongoing Practice

The question “is this food?” is not answered once. It is answered continuously, across every intake decision, across every substance the reader encounters. The paper installs a practice; the practice is applied across a lifetime.

For each substance the reader considers eating, they ask the four questions and place the substance in a category. They notice when their cultural framing and the body-response framing diverge, and they go with body response. They notice when a substance they had been treating as regular food is actually irregular food at the frequency they’re eating it. They notice when a substance they had been avoiding as “bad” is actually fine at occasional intake.

Over time, the practice becomes automatic. The reader knows, without thinking through the rule explicitly, where most substances fall for them. The practice produces a calibrated intuition that tracks the body’s actual response, rather than the cultural framing that tracks whatever framings happen to be in circulation.

The paper is not a list. The reader who wants a list of approved foods and forbidden foods has not received what the paper provides. The paper provides a method. The reader applies the method. The results of the method are specific to the reader’s body, intake pattern, and context.

15. What You Can Now Do

If you have read this paper as its target reader and the transmission worked, you now have the following:

You can distinguish food from non-food essentials, poisons, homeostasis-disruptors at excess dose, and medicines. These are not different styles of food; they are different categories with different relationships to the body.

You understand the body as a homeostasis machine running continuous regulation, and you understand food as an input to that regulation. You ask what substances do to the machine rather than what they are made of.

You evaluate substances by body response over time, not by cultural category or chemical composition. You know that the chemistry is upstream and the response is the measurement.

You place substances on the sustaining gradient. You recognize that pepper seeds are insufficient, that rabbit meat alone is acutely non-sustaining, that pig meat broadly sustains, that no single food is perfectly complete for a lifetime alone.

You account for duration and frequency in food status. You recognize the short-term-food, long-term-poison category as real, and you understand that many processed foods live there.

You look past food names to specific substances. You know that “bread” is a range and the body responds to the specific bread, not to the category.

You apply the regular/irregular split to intake decisions. You know that the split is defined by body response, not by cultural custom, and you use it to decide what to eat as the basis of your diet versus what to eat occasionally.

You walk through the four-question decision rule for any substance you encounter. You arrive at a stable placement that tracks what the body will do with the substance over the intake pattern you’re considering.

You recognize when a substance you had been treating one way is actually something else by body response, and you adjust.

This is the post-read state the paper was written to produce. If you have it, the transmission worked. If you don’t, some platform in the paper failed for you, and the failure is diagnostic: go back to the platform that didn’t land, work through it until it does, then continue.

16. Closing

Food is what sustains the body’s homeostasis machine when ingested. Not-food is what destabilizes it. Most substances do both in varying proportions, and the category of any given substance depends on the net regulatory effect over the intake pattern.

This is a simple definition. It is also a demanding one, because it forces the reader to evaluate substances by what the body actually does with them rather than by what the substance is called, what culture says about it, or what advertising associates with it. The body’s response is observable. The response accumulates over years into measurable health outcomes. The responses are what actually determine whether a person thrives or declines under their intake pattern.

The modern food environment is dominated by substances that are irregular food by body response. The body’s regulatory machinery responds to substances as the substances are, not as they are packaged. A person who eats regularly from the irregular-food pool produces the regulatory wear that corresponds to that pattern, regardless of whether the pool is culturally positioned as normal food. The regulatory wear appears as the chronic disease profile that dominates modern mortality.

The method in this paper is not a diet. It is not a prescription. It is a way of asking “is this food?” that produces answers corresponding to what the body will actually do with the substance. The reader applies the method to their own intake, in their own context, with their own body’s particular tolerances. The results are specific to them. The method is general.

The practice is what matters. Reading this paper installs the frame. Applying it to real intake decisions, across years, produces a person whose eating tracks their body’s actual response rather than the cultural categories they happen to have inherited. Over a lifetime, the difference between eating by body response and eating by cultural category is the difference between a regulatory machine that runs well for decades and a regulatory machine that wears down under inputs it was not designed to handle.

You ask “is this food?” for every substance you encounter. The four questions give you the answer. The answer tracks your body. Your body’s response, over years, is what determines the health you live in. The method is installed. The practice is yours.

References

[[HOWL-BODY-2-2026](#)] What is Food?. Github: <https://github.com/ghowland/papers/tree/main/papers/BODY/HOWL-BODY-2-2026>

[[HOWL-BODY-1-2026](#)] Neural Territory. Github: <https://github.com/ghowland/papers/tree/main/papers/BODY/HOWL-BODY-1-2026>

Appendix: Supporting Reference Tables

Table A1: The Category Taxonomy

Every ingested substance falls into one of these categories. The category is determined by the substance’s relationship to the body, not by properties of the substance in isolation.

Category	Definition	Body Response	Examples
Poison	Harms at any dose. No intake level helps the body sustain function.	Cellular or metabolic injury proportional to dose.	Cyanide, arsenic, botulinum toxin, mercury, lead
Non-food essential	Required for life, ingested or inhaled, cannot sustain life alone.	Regulatory function supported; no caloric substrate provided.	Water, air, salt

Category	Definition	Body Response	Examples
Homeostasis-disruptor at excess dose	Safe or required at normal doses, lethal when dose overwhelms regulatory capacity.	Regulatory systems overwhelmed by volume, not by chemical injury.	Water past LD50, oxygen at high pressure, sodium at extreme intake
Medicine	Situational intervention. Functions as medicine only at correct dose with correct indication.	Specific effect produced through mechanism; intervention in function rather than sustaining of function.	NSAIDs with inflammation, antibiotics with infection, insulin with diabetes
Insufficient substance	Ingestible, not toxic, provides too little for the body to extract meaningful sustenance.	Approximately equivalent to no intake in terms of sustenance.	Pepper seeds, most spices alone, inedible fiber
Partial food — acutely non-sustaining	Real food, but fails as sole intake within weeks through specific mechanisms.	Kills through metabolic overload or deficit before pure starvation timeline.	Rabbit meat alone, very lean fish alone
Partial food — acutely sustaining, chronically insufficient	Provides calories and prevents acute starvation, develops deficiency disease over months to years.	Survival short-term, progressive deficiency and regulatory wear.	Bread and water, plain polished rice alone
Sustaining food with chronic costs	Can sustain life long-term as sole intake, with specific slowly-developing problems.	Maintains basic homeostasis for years; specific systems accumulate stress.	Pig meat alone, potato-dominant diets
Regular food	Body sustains indefinitely on this as ongoing intake.	Regulation succeeds, homeostasis maintained, no accumulating load.	Varied whole foods: meats, eggs, vegetables, fruits, whole grains, legumes, nuts
Irregular food	Body handles occasionally without lasting harm; frequent intake destabilizes.	Transient perturbation at occasional intake; regulatory wear at sustained intake.	Cake, chips, sweetened drinks, fried foods, alcohol, ice cream

Category	Definition	Body Response	Examples
Short-term food, long-term poison	Passes acute food test, fails chronic food test.	Sustains brief exposure, actively degrades regulatory machine under sustained intake.	Processed flour, heavily processed foods, industrial seed oils

Table A2: The Sustaining Gradient

Within the category of food, substances vary in how completely they sustain the body as sole intake. This table places example substances on the gradient.

Position	Substance	Time to Failure as Sole Intake	Primary Failure Mechanism	Status
Below threshold	Pepper seeds, spices alone	~50 days	Insufficiency equivalent to starvation	Not food
Acutely non-sustaining	Rabbit meat alone	2–4 weeks	Nitrogen toxicity from excess protein without fat	Partial food, fails alone
Acutely non-sustaining	Pure sugar and water	3–8 weeks	Broad regulatory failure; no protein, fat, or micronutrients	Irregular food, near poison
Acutely sustaining, chronically insufficient	White bread and water	3–12 months	Deficiency disease: scurvy, B-vitamin deficiencies, essential fatty acid deficiency	Partial food, inadequate alone
Acutely sustaining, chronically insufficient	Polished rice and water	3–12 months	Beriberi (thiamine deficiency), protein insufficiency	Partial food, inadequate alone

Position	Substance	Time to Failure as Sole Intake	Primary Failure Mechanism	Status
Broadly sustaining	Pig meat alone	Years	Gout, cardiovascular effects, vitamin C deficiency over long time	Sustaining food with costs
Broadly sustaining	Whole milk for adapted populations	Years	Specific deficiencies over time; works best for some populations	Sustaining food with costs
Broadly sustaining	Eggs as primary food	Years	Specific deficiencies very slowly	Sustaining food with costs
Varied intake	Combined whole foods (meat, eggs, vegetables, fruits, grains, dairy)	Indefinite	None at reasonable variation	Regular food

Table A3: Intake Frequency and Food Status

The same substance can change category based on how often it is eaten. This table shows how frequency determines category for representative substances.

Substance	Rare (monthly or less)	Occasional (weekly)	Frequent (daily)	Dominant (major caloric fraction)
Home-made whole wheat bread	Regular food	Regular food	Regular food	Partial food; combine with other intake
White bread	Regular food	Borderline regular/irregular	Irregular food	Chronically insufficient
Cheap white bread	Irregular food	Irregular food	Chronically degrading	Chronically degrading with active damage
Cheap potato chips	Irregular food	Irregular food	Chronically degrading	Rapidly degrading

Substance	Rare (monthly or less)	Occasional (weekly)	Frequent (daily)	Dominant (major caloric fraction)
Cake (typical recipe)	Regular-equivalent for the occasion	Irregular food	Chronically degrading	Severely destabilizing
Pure sugar (added)	Irregular food	Irregular food	Chronically degrading	Near-poison level destabilization
Sweetened soft drinks	Irregular food	Chronically degrading	Severely destabilizing	Approaching pure-sugar failure
Pig meat	Regular food	Regular food	Regular food	Sustaining food with chronic costs
Eggs	Regular food	Regular food	Regular food	Broadly sustaining
Whole fruit	Regular food	Regular food	Regular food	Partial food; combine with other intake
Fruit juice (added sugar)	Irregular food	Irregular food	Chronically degrading	Severely destabilizing
Alcohol (moderate portion)	Irregular food	Irregular food	Chronically degrading	Severely destabilizing

Table A4: Regulatory Systems and Food Impact

The body's major regulatory systems, what they require from food to function, and what destabilizes them.

Regulatory System	What It Regulates	What It Requires from Food	What Destabilizes It
Blood glucose regulation	Serum glucose, insulin, glucagon	Variable carbohydrate intake, fiber, protein, fat to slow absorption	Frequent high-glycemic loads, isolated sugar, processed flour
Electrolyte balance	Sodium, potassium, chloride, magnesium, calcium	Varied intake of minerals, adequate water	Extreme salt intake, potassium deficit, dehydration
Fluid regulation	Blood volume, cellular hydration, osmotic pressure	Water, sodium in range, potassium	Excess sodium, insufficient water, excess water

Regulatory System	What It Regulates	What It Requires from Food	What Destabilizes It
Acid-base regulation	Blood pH, buffer systems	Varied intake contributing balanced acid/base load	Extreme protein loads, acidic processed foods in dominance
Protein and nitrogen regulation	Amino acid pool, urea cycle, muscle maintenance	Complete protein in adequate but not excessive amounts	Protein deficit (muscle wasting), protein excess without fat (nitrogen toxicity)
Fat regulation	Essential fatty acids, cell membranes, hormone synthesis	Dietary fats including essential fatty acids	Insufficient fat, excess oxidized fats, imbalanced omega-3/omega-6
Micronutrient regulation	Enzyme cofactors, vitamin-dependent processes	Vitamins and minerals from varied whole foods	Monotonous intake lacking specific nutrients, malabsorption
Gut regulation	Microbiome, gut barrier, nutrient absorption	Fiber, varied substrates, fermented foods in some traditions	Processed foods lacking fiber, excess sugar, additive load
Immune regulation	Inflammatory response, pathogen defense	Adequate protein, zinc, vitamin C, vitamin D, varied inputs	Chronic inflammation from processed foods, deficiency states
Cardiovascular regulation	Blood pressure, vascular integrity	Balanced electrolytes, adequate nutrients for vessel maintenance	Excess sodium, oxidized fats, chronic high-glycemic load, deficiency in specific minerals
Endocrine regulation	Hormone synthesis and clearance	Cholesterol, amino acids, fat-soluble vitamins, specific minerals	Insufficient fat, excess sugar, specific nutrient deficiencies
Liver regulation	Detoxification, metabolic processing	Manageable load of substances to clear	Alcohol excess, pharmaceutical load, additive-heavy diet

Table A5: Failure Timelines for Sole Intake

Approximate time to significant dysfunction or death when a single substance is the only intake. These are approximations from historical and experimental data.

Substance	Time to Weakness	Time to Severe Dysfunction	Time to Death	Primary Lethal Mechanism
Nothing (water only)	2–3 weeks	3–5 weeks	~50 days	Starvation
Salt (with water)	Hours	1–2 days	3–7 days	Dehydration, cardiovascular collapse
Pepper seeds (with water)	2–3 weeks	3–5 weeks	~50 days	Starvation (functionally equivalent to no food)
Rabbit meat alone	1–2 weeks	3–4 weeks	4–8 weeks	Nitrogen toxicity, caloric insufficiency
Pure sugar (with water)	2–4 weeks	4–8 weeks	2–4 months	Multi-system regulatory failure
White bread (with water)	1–2 months	3–6 months	6–18 months	Deficiency disease (specific to what's missing)
Polished rice (with water)	1–2 months	3–6 months	6–12 months	Beriberi, protein insufficiency
Potatoes alone (whole)	Months	1–2 years	Years	Slow deficiency (particularly protein, specific micronutrients)
Pig meat alone	Years	Years to decades	Decades	Gout, cardiovascular, eventual specific deficiencies

Substance	Time to Weakness	Time to Severe Dysfunction	Time to Death	Primary Lethal Mechanism
Eggs alone	Years	Decades	Very slow	Minimal specific deficiencies over very long time
Whole milk alone (for adapted populations)	Years	Decades	Very slow	Iron deficiency, specific issues over long time
Varied whole foods	Indefinite	Indefinite	Normal lifespan	Not sole-intake failure

Table A6: The Four-Question Decision Rule

The operational rule for evaluating any substance against “is this food?”

Question	What It Assesses	Substance Passes If	Substance Fails If
1. What does this substance provide the homeostasis machine?	Contribution of energy substrate, structural materials, and regulatory materials	Provides meaningful amounts of multiple categories (calories plus protein, or calories plus micronutrients, or ideally all three)	Provides only calories, or nothing meaningful
2. What load does this substance put on the regulatory systems?	Stress on specific regulatory pathways	Load is within normal capacity of the systems involved	Strains specific systems (insulin, kidney, liver, electrolyte) beyond comfortable range
3. What happens under sustained intake?	Whether the body maintains regulation over time	Regulation remains stable; no specific systems degrade progressively	Regulatory capacity degrades; specific systems wear down; chronic disease develops

Question	What It Assesses	Substance Passes If	Substance Fails If
4. What happens under occasional intake?	Whether the body recovers from periodic exposure	Body absorbs, processes, and returns to baseline without lasting effect	Damage accumulates even at occasional exposure; substance is poison or homeostasis-disruptor

Category Assignment from Question Results

Pass 1	Pass 2	Pass 3	Pass 4	Category
Yes	Yes	Yes	Yes	Regular food
Partial	Yes	Partial	Yes	Partial food — regular in combination
Yes	Partial	No	Yes	Short-term food, long-term poison / Irregular food
Minimal	Partial	No	Yes	Irregular food
Minimal	No	No	No	Homeostasis-disruptor / Near-poison
No	No	No	No	Poison
No (not substrate)	Yes (required)	N/A	N/A	Non-food essential
Varies	Varies	Moot	Moot	Medicine (situational)

Table A7: Substances Arrayed on the Full Spectrum

This table places example substances on the spectrum from most sustaining to most destabilizing. The ordering is approximate, and position depends on intake pattern.

Position	Substance	Category	Dominant Body Response
Most sustaining	Eggs	Regular food	Broad stabilization across most systems

Position	Substance	Category	Dominant Body Response
	Organ meats in variety	Regular food	Dense micronutrient stabilization
	Fatty fish (wild-caught)	Regular food	Stabilization plus essential fatty acids
	Varied meats and seafood	Regular food	Protein, fat, micronutrients
	Varied vegetables	Regular food	Micronutrients, fiber; partial alone
	Whole fruits in variety	Regular food	Micronutrients, sugar in fiber matrix
	Whole grains	Regular food in combination	Carbohydrate, some protein, fiber, micronutrients
	Legumes	Regular food in combination	Protein, carbohydrate, fiber, micronutrients
	Nuts and seeds	Regular food	Fat, protein, micronutrients
	Raw/lightly processed dairy (for tolerant populations)	Regular food	Fat, protein, calcium, other micronutrients
	Home-made whole wheat bread	Partial food	Like whole grains, less dense than original grain
	Potatoes (whole)	Partial food	Carbohydrate, potassium, some protein
	Store whole wheat bread	Partial food toward irregular	Reduced nutrition, some additive load
	White bread (enriched)	Borderline regular/irregular	Calories with minimum nutrient profile
	Cheap white bread	Irregular food	Calories plus additive load, minimum nutrients
	Sweetened yogurt (commercial)	Irregular food	Some dairy benefit plus sugar load
	Commercial breakfast cereals	Irregular food	Processed grain plus concentrated sugar

Position	Substance	Category	Dominant Body Response
	Cake (typical)	Irregular food	Concentrated sugar plus processed flour plus fat
	Cheap potato chips	Irregular food toward poison	Oxidized fats, sodium, minimal nutrition
	Fast food hamburger	Irregular food	Variable; depends on specific product
	Ice cream	Irregular food	Sugar, fat, some dairy benefit
	Sweetened soft drinks	Irregular food near pure sugar	Concentrated sugar in water
	Pure sugar	Irregular food near poison	Calories only, severe regulatory load
	Alcohol	Irregular food near poison	Calories with hepatic clearance load and neurological effect
	Fruit juice (sweetened)	Irregular food near pure sugar	Concentrated sugar without fiber
	Industrial seed oils (used repeatedly)	Short-term food, long-term poison	Calories with oxidized fat load
	Deep-fried processed foods	Short-term food, long-term poison	Calories with multiple destabilizers
Not food (essentials)	Water	Non-food essential	Hydration support
	Salt	Non-food essential	Electrolyte support
	Air	Non-food essential	Oxygen for respiration
Not food (harmful)	Small doses of mild poisons	Poison at any dose	Proportional injury
	Cyanide	Poison	Cellular respiration arrest
Most destabilizing	Acute lethal poisons	Poison	Rapid death

Table A8: Common Modern Foods Versus Cultural Framing

Substances often positioned as regular food that are irregular food by body response, and substances sometimes avoided that are actually regular food.

Substance	Cultural Position	Body-Response Category	Reason for Mismatch
Commercial breakfast cereals (sweetened)	“Healthy breakfast food”	Irregular food	Sugar content closer to dessert than to whole grain
Fruit juice	“Healthy beverage”	Irregular food	Concentrated sugar without fiber of whole fruit
Flavored yogurt	“Healthy snack”	Irregular food	Added sugar often exceeds beneficial dairy content
Granola bars (commercial)	“Healthy snack”	Often irregular food	Typically high in added sugar and processed oils
Smoothies (sweetened commercial)	“Healthy drink”	Often irregular food	Concentrated sugar with variable real content
Sports drinks	“Healthy hydration”	Irregular food	Sugar and sodium solution; not required for normal activity
Diet sodas	“Better than regular soda”	Non-food; irregular at best	Artificial sweeteners have their own regulatory effects
Whole eggs	Sometimes avoided for cholesterol	Regular food	Broad nutritional content; cholesterol concerns largely overstated
Full-fat dairy (for tolerant)	Sometimes avoided for fat content	Regular food	Fat content is a feature; fat-soluble vitamins require fat
Red meat	Sometimes positioned as irregular	Regular food for most people	Broad nutritional content; concerns context-dependent
Saturated fats from whole sources	Sometimes avoided	Regular food component	Required for hormone synthesis and cell membranes
Salt (in normal cooking amounts)	Sometimes over-restricted	Non-food essential, use as needed	Required for regulatory function; deficiency is real problem

Table A9: Duration-Dependent Substances — The Short-Term-Food, Long-Term-Poison Profile

Substances that pass the acute food test and fail the chronic food test. Each has a characteristic timeline for transition from food-like to destabilizing at sustained intake.

Substance	Acute Food Status	Chronic Status (Frequent Intake)	Primary Destabilization Mechanism	Approximate Transition Timeline
Processed flour	Food (provides calories, brief sustenance)	Active destabilizer	Glycemic load, micronutrient deficit, gut effects	Months to years of dominant intake
Industrial seed oils	Food (calories)	Active destabilizer	Oxidized fats, omega imbalance, inflammation	Months to years of regular use
High-fructose corn syrup	Food (calories)	Active destabilizer	Hepatic load, glucose regulation strain	Weeks to months at significant intake
Deep-fried foods	Food (calories)	Active destabilizer	Oxidized fats plus processing effects	Months to years of regular intake
Processed meat (with nitrites, additives)	Food (protein, fat)	Slow destabilizer	Preservative load, altered protein matrix	Years of regular intake
Alcohol	Tolerable at small doses	Active destabilizer	Hepatic load, neurological effects, caloric displacement	Months to years of regular intake
Artificially sweetened products	Minimal food value	Variable destabilizer	Microbiome effects, possible metabolic signaling issues	Months to years
Ultra-processed foods (composite)	Food (calories)	Active destabilizer	Combined effects of above mechanisms	Months to years

Table A10: Individual Variation Factors

Food status is generally consistent across individuals, but specific tolerances vary. This table names the major axes of individual variation that affect where substances fall for specific people.

Variation Axis	What It Affects	Common Examples
Lactase persistence	Dairy tolerance	Fresh milk: regular food for many populations, irregular or non-food for others
Gluten sensitivity or celiac disease	Wheat tolerance	Wheat: regular food for most, non-food (harmful) for celiac sufferers
FODMAP sensitivity	Gut response to certain carbohydrates	Onions, garlic, legumes: regular for most, irregular for sensitive individuals
Specific food allergies	Immune response to specific proteins	Nuts, shellfish, eggs: regular food for most, poison for allergic individuals
Enzyme variations	Processing of specific substances	Alcohol tolerance varies by genotype; some cannot safely consume
Metabolic conditions	Handling of specific macronutrients	Diabetics handle glucose differently; PKU sufferers cannot handle phenylalanine
Age-related changes	Nutritional needs and tolerance	Elderly often need more protein; children need specific nutrient densities
Pregnancy and lactation	Specific increased needs	Folate, iron, calcium needs higher; some substances (alcohol) shift toward poison
Activity level	Caloric and specific nutrient needs	High-activity: more calories, specific electrolytes; sedentary: less
Gut microbiome composition	Processing of various foods	Individual microbiomes handle fiber and fermentable substances differently
Historical adaptation	Population-level food tolerances	Inuit populations adapted to high-fat marine diets; others less so

Variation Axis	What It Affects	Common Examples
Acquired sensitivities	Changed tolerance over time	Gut damage from previous illness can reduce tolerance for previously-regular foods

Table A11: The Decision Rule in Summary

A condensed reference for applying the method.

Step	Question	Answer	Action
1	What does this provide?	Meaningful substrate, structure, and regulatory materials	Proceed to step 2 with positive weight
1	What does this provide?	Only calories or less	Proceed to step 2 with negative weight
2	What load does it impose?	Within regulatory capacity	Proceed to step 3 with positive weight
2	What load does it impose?	Strains specific systems	Proceed to step 3 with negative weight
3	What happens under sustained intake?	Body sustains	Regular food (at this intake pattern)
3	What happens under sustained intake?	Regulatory wear accumulates	Step 4 determines whether irregular food or worse
4	What happens under occasional intake?	Body absorbs and recovers	Irregular food
4	What happens under occasional intake?	Damage accumulates even occasionally	Not food — poison or disruptor

Practical Application Heuristic

Observation	Likely Category
Traditional food eaten for centuries by populations in good health	Regular food for those populations
Food requiring significant processing, additives, and engineering	Likely irregular food or short-term food, long-term poison

Observation	Likely Category
Food that people can eat but that makes them feel worse over time	Irregular food at that frequency, or not tolerated by that individual
Food that concentrates specific substances (sugar, sodium, oils) beyond natural levels	Irregular food; may be short-term food, long-term poison
Food sold primarily for convenience or flavor engineering	Often irregular food regardless of marketing
Substance required for regulatory function but not providing substrate	Non-food essential
Substance harmful at any dose	Poison
Substance producing a specific effect for a limited purpose	Medicine (situational)